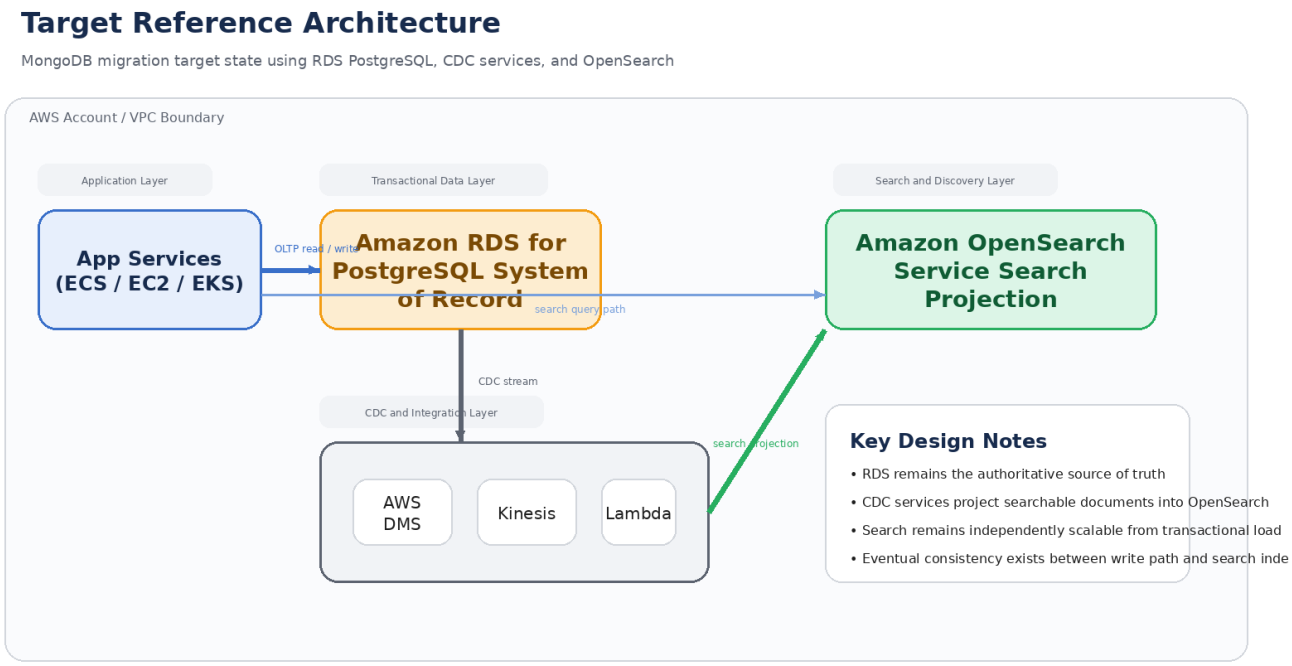
ADR-001: MongoDB to AWS Migration

Boardroom-grade ADR with updated visuals and directional cost framing

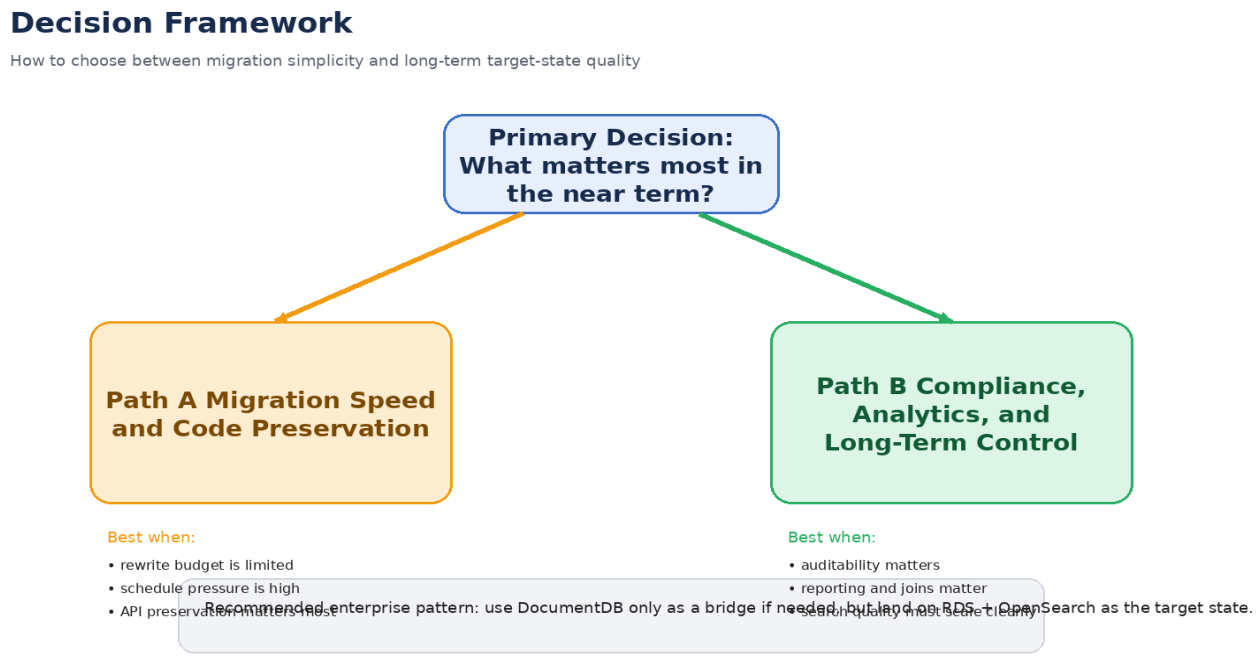
Executive Summary

This ADR recommends migrating MongoDB workloads to an AWS-native polyglot architecture using **Amazon RDS for PostgreSQL** as the system of record and **Amazon OpenSearch Service** as the search layer. **Amazon DocumentDB** may be used as a transitional bridge when migration simplification is the dominant near-term requirement.

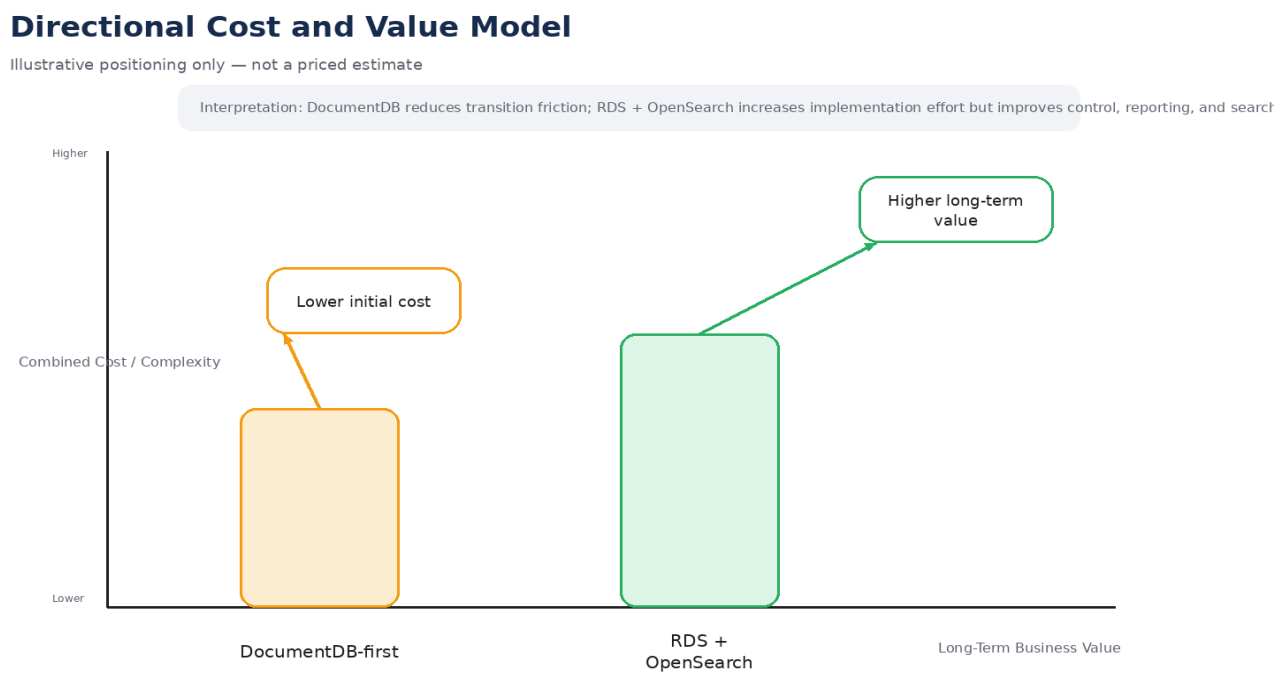
Target Reference Architecture



Decision Framework



Directional Cost and Value Model



Recommendation

Track	Position

Target state	RDS PostgreSQL + OpenSearch
Bridge option	DocumentDB only when migration friction is the primary concern

Migration Roadmap

Phase	Description
Assess	Inventory MongoDB clusters, collections, query patterns, and search usage.
Design	Define the RDS target model, OpenSearch index model, and CDC path.
Build	Stand up RDS, CDC services, monitoring, and OpenSearch.
Migrate	Run bulk migration plus incremental synchronization.
Refactor	Update application reads, writes, and search calls.
Cut over	Switch traffic, verify reconciliation, and retire MongoDB.